

Linear systems of three unknowns



1 Determine whether each ordered triple is a solution of the corresponding system of equations:

a Triple: $(-9, -7, 4)$

System:

$$-2x + 5y - 3z = -29$$

$$-3x + y + 4z = 38$$

$$-4x + 2y + 3z = 34$$

b Triple: $(1, -5, -3)$

System:

$$4x - 5y + 3z = 20$$

$$4x - 5y - z = 32$$

$$-x + 2y - 5z = 4$$

c Triple: $\left(\frac{3}{4}, \frac{4}{5}, \frac{2}{5}\right)$

System:

$$5x + y + 3z = \frac{23}{4}$$

$$4x + 3y + 5z = \frac{37}{5}$$

$$2x - 4y + 3z = -\frac{1}{2}$$

d Triple: $(1.1, 0.3, 1.7)$

System:

$$3x + y - 5z = -5.9$$

$$5x + y - 4z = -1$$

$$5x - 2y - 4z = -1.9$$

e Triple: $(-8, -1, 2)$

System:

$$x + 5y + 4z = -8$$

$$4x - y + 3z = -25$$

$$2y - z = -4$$

f Triple: $(-6, 3, -9)$

System:

$$x + 5y + 3z = -18$$

$$5x + y - 3z = -3$$

$$5x + 3y - z = -12$$

2 Solve the following systems of equations:

a $x - 5y + z = -16$

$$2x + 5y + 2z = 13$$

$$-x - 5y - 3z = -10$$

b $x + y = 6$

$$y + z = 13$$

$$x + z = 9$$

c $x + 3y = -11$

$$5y + 4z = -33$$

$$x + 2z = 0$$

d $5x + 2y + 4z = -13$

$$6y + 4z = 22$$

$$5x - 2y + 3z = -31$$

e $3x + 2y - z = -9$

$$6x - 4y + z = -20$$

$$3x + 6y + 4z = 5$$

f $3x + 3y - z = 6$

$$2x - 2y + z = 3$$

$$2x + 5y - z = 2$$

g $3x + 2y - 2z = 4$

$$y + 2z = 5$$

$$3x - y - 2z = 7$$

h $x = 8(y - z)$

$$2z = 9(y - x)$$

$$x + z = 2y - 3$$

i $x = 2(y - z)$

$$y = 4(z - x)$$

$$x + y = z + 6$$

j $9x + 5y - 2z = 25$

$$x + 4y - 7z = 30$$

$$3x - 6y + 8z = -33$$

3 How many solutions do each of the following systems of equations have?



a
$$\begin{aligned} 3x + 4y + 2z &= -1 \\ 5x - 2y - 3z &= -23 \\ -3x - 4y - 2z &= 1 \end{aligned}$$

b
$$\begin{aligned} 6x + 7y - 4z &= -13 \\ 12x + 14y - 8z &= -24 \\ 3x - 2y + 5z &= 17 \end{aligned}$$

4 Consider the system:

Equation 1	Equation 2
$2x - 3y - z = -19$	$y - z = -1$

Let $z = t$ be arbitrary.

- a Solve Equation 2 for y with respect to t .
- b Using your result from part a, solve Equation 1 for x with respect to t .
- c Hence, write down the solution to the system with respect to the arbitrary variable t .

5 Consider the system:

Equation 1	Equation 2
$4x + 5y - z = 11$	$x + y + 2z = 9$

Let $z = t$ be arbitrary.

- a Solve Equation 2 for y with respect to x and t .
- b Solve Equation 1 for x with respect to t .
- c Hence, solve for y with respect to t .
- d Hence, write down the solution to the system with respect to the arbitrary variable t .

6 Consider the system:

Equation 1	Equation 2
$x - 3y + 3z = 6$	$x - 2y + 4z = 5$

Let $z = t$ be arbitrary.

- a Solve Equation 1 for x with respect to y and t .
- b Solve Equation 2 for y with respect to t .
- c Hence, solve for x with respect to t .
- d Hence, write down the solution to the system with respect to the arbitrary variable t .

- 7 Consider the following system of three equations:



$$\begin{aligned}3x + 5y + 8z &= 136 \\27x + 45y + 72z &= 872 \\x + y + z &= 25\end{aligned}$$

- a Does the system have one solution, infinite solution or no solution?
- b Is the system consistent or inconsistent?

- 8 Consider the following system of 3 equations:

$$\begin{aligned}8x + 5y + 8z &= 97 \\6x + y + 4z &= 45 \\x + y + z &= 14\end{aligned}$$

- a Solve the system.
- b Is the system independent or dependent?

- 9 Consider the following system of 3 equations:

$$\begin{aligned}7x + 3y + 3z &= 3.2 \\8x + 5y + 3z &= 4.4 \\4x + 9y + 4z &= 5.7\end{aligned}$$

- a Solve the system.
- b Is the system consistent or inconsistent?

- 10 Find a system of three linear equations in three variables that has $(-4, -2, -1)$ as a solution.

- 11 Solve the following systems of equations:

a
$$\begin{aligned}\frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= 13 \\ \frac{1}{x} + \frac{1}{y} &= 9 \\ \frac{1}{x} - \frac{1}{z} &= 2\end{aligned}$$

b
$$\begin{aligned}\frac{1}{x} + \frac{1}{y} + \frac{1}{z} &= 11 \\ \frac{1}{y} + \frac{1}{z} &= 8 \\ \frac{1}{x} - \frac{1}{y} &= -3\end{aligned}$$

12 For each of the following system of 3 equations:



- i Solve the system.
- ii Determine whether the system is consistent or inconsistent.
- iii Determine whether the system is dependent or independent.

a $x - 5z = -8$
 $y + 4z = 16$
 $6y + 2z = 30$

b $x + 3z = 22$
 $y + 8z = 49$
 $8z = 48$

c $x + 5y - 7z = -8$
 $y + 6z = 56$
 $6y + 9z = 120$

d $9x + 9y + 6z = 303$
 $16x + 18y + 10z = 536$
 $x + y + z = 39$

e $5x + 4z = 42$
 $5x + 9y = 37$
 $6y + 7z = 74$

f $9x + 7y = 180$
 $x + y + 5z = 27$
 $4x + 6y + 4z = 110$

g $9x + 7y + 3z = 151$
 $8x + y + 9z = 142$
 $x + y + z = 23$

h $8x + 9y + 2z = 51$
 $4x + 8y + 9z = 101$
 $8x + 4y + 9z = 109$

i $3x + 7y = 4.4$
 $x + y + 6z = 3.8$
 $5x + 6y + 8z = 8.5$

Applications

13 Consider the equation: $Ax + By + Cz = 12$.

- a One solution to this equation is $\left(1, \frac{3}{4}, 3\right)$. Substitute these values into the equation, to give an equation in terms of A , B , and C .
- b A second solution to the equation is $\left(\frac{4}{3}, 1, 2\right)$. Substitute these values into the equation, to give an equation in terms of A , B , and C .
- c A third solution to the equation is $(2, 1, 1)$. Substitute these values into the equation, to give an equation in terms of A , B , and C .
- d Solve your three equations for A , B , and C .

- 14 The fraction $\frac{1}{24}$ can be written as the following sum:

$$\frac{1}{24} = \frac{x}{8} + \frac{y}{4} + \frac{z}{3}$$

where the numbers x , y , and z are solutions of:

$$x + y + z = 1 \text{ --- equation 1}$$

$$2x - y + z = 0 \text{ --- equation 2}$$

$$-x + 2y + 2z = -1 \text{ --- equation 3}$$

- Solve this system of equations for x , y , and z .
- Use your values of x , y , and z to determine whether the following equation is true or false:

$$\frac{1}{24} = \frac{x}{8} + \frac{y}{4} + \frac{z}{3}$$

- 15 Consider the quadratic with equation $y = ax^2 + bx + c$ that passes through the points $(1, 15)$, $(2, 5)$ and $(4, -3)$.

- Set up a system of equations in the variables a , b and c using the points given.
- Solve your equations for a , b , and c .
- Write down the quadratic equation that passes through the points $(1, 15)$, $(2, 5)$ and $(4, -3)$.

- 16 The equation of a circle can be written in the general form $x^2 + y^2 + ax + by + c = 0$. The circle below passes through the points $(2, -4)$, $(5, 1)$ and $(3, -3)$:

- The general form of a circle given may be rearranged to the following form: $ax + by + c = -(x^2 + y^2)$. Use this form to set up a system of equations in the variables a , b and c from the given points.
- Solve your equations for a , b , and c .
- Write down the equation of the circle that passes through the points $(2, -4)$, $(5, 1)$ and $(3, -3)$.

